

a] Write a shell script to generate mark-sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.

Code: echo "Enter marks of
English" read m1 echo "Enter
marks of Mathematics" read m2
echo "Enter marks of Science" read
m3 total=\$((m1+m2+m3))
per=\$((total/3)) echo "Student
Total Marks=\$total" echo
"percentage=\$per" if [\$per -gt 75
]; then
 echo "Class: Distinction"
elif [\$per -gt 60]; then
 echo "class: First Class"
elif [\$per -gt 40]; then
 echo "class: Second Class"
elif [\$per -gt 35]; then
 echo Class: Third Class"
else
 echo "class: Fail"
fi

OUTPUT:

```
MINGW64:/c:/Users/Hp/Desktop/OSCD25D008
Hp@89L1D0M0F305T MINGW64 ~/Desktop (master)
$ cd OSCD25D008
Hp@89L1D0M0F305T MINGW64 ~/Desktop/OSCD25D008 (main)
$ echo "Enter marks of English"
read m1
echo "Enter marks of Mathematics"
read m2
echo "Enter marks of Science"
read m3
total=$((m1 + m2 + m3))
per=$((total / 3))
echo "Student Total Marks = $total"
echo "Percentage = $per"
if [ $per -gt 75 ]; then
    echo "Class: Distinction"
elif [ $per -gt 60 ]; then
    echo "Class: First Class"
elif [ $per -gt 40 ]; then
    echo "Class: Second Class"
elif [ $per -gt 35 ]; then
    echo "Class: Third Class"
else
    echo "Class: Fail"
fi
Enter marks of English
76
Enter marks of Mathematics
87
Enter marks of Science
98
Student Total Marks = 261
Percentage = 87
Class: Distinction
```

b) Write a menu driven shell script which will print the following menu and execute the given task.

- Display calendar of current month.
- Display today's date and time.
- Display usernames those are currently logged in the system.
- Display your terminal number.

CODE:

```
#!/bin/bash
```

```
echo "1. Display current month and year"
```

```
echo "2. Display today's date and time"
```

```
echo "3. Display logged in users" echo "4.
```

```
Display terminal number" echo "Enter
your choice:" read choice
```

```
case $choice in
```

```
1) date +"%B %Y" ;;
```

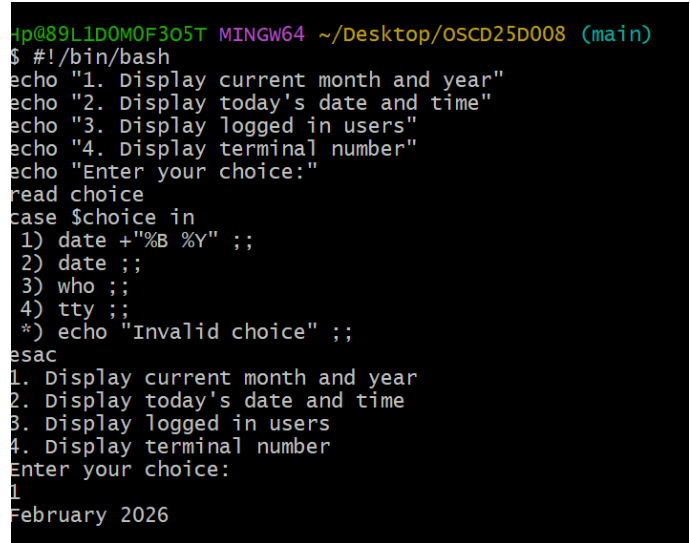
```
2) date ;;
```

```
3) who ;;
```

4) tty ;;

*) echo "Invalid choice" ;; esac

OUTPUT:



```
hp@89L1D0M0F305T MINGW64 ~/Desktop/OSCD25D008 (main)
$ #!/bin/bash
echo "1. Display current month and year"
echo "2. Display today's date and time"
echo "3. Display logged in users"
echo "4. Display terminal number"
echo "Enter your choice:"
read choice
case $choice in
  1) date +"%B %Y" ;;
  2) date ;;
  3) who ;;
  4) tty ;;
  *) echo "Invalid choice" ;;
esac
1. Display current month and year
2. Display today's date and time
3. Display logged in users
4. Display terminal number
Enter your choice:
1
February 2026
```

C] Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13

CODE: echo "Enter number of

terms:" read n a=1 b=1 echo

"Fibonacci Series:"

for ((i=1; i<=n; i++))

do echo -n

"\$a " c=\$((a

+ b)) a=\$b

b=\$c

done echo

OUTPUT:

```

Hp@89L1D0M0F3O5T MINGW64 ~/Desktop/OSCD25D008 (main)
$ echo "Enter number of terms:"
read n
a=1
b=1
echo "Fibonacci Series:"
for (( i=1; i<=n; i++ ))
do
    echo -n "$a "
    c=$((a + b))
    a=$b
    b=$c
done
echo
Enter number of terms:
2
Fibonacci Series:
1 1

```

D] Write a shell script which will accept a number n and display first n prime numbers as output.

CODE:

```

echo "Value for n :
" read n count=0

num=2

echo "Display of first n prime numbers"
echo "First $n prime numbers are:"

while [ $count -lt $n ]
do
    flag=0    for (( i=2;
i<=num/2; i++ ))    do

        if [ $((num % i)) -eq 0 ]; then

            flag=1
break
        fi

```

```

done    if [ $flag -eq 0
]; then    echo -n
"$num "
count=$((count + 1))
fi
num=$((num + 1))
done echo

```

OUTPUT:

```

Hp@89L1D0M0F3O5T MINGW64 ~/Desktop/OSCD25D008 (main)
$ echo "Value for n : "
read n
count=0
num=2
echo "Display of first n prime numbers"
echo "First $n prime numbers are:"
while [ $count -lt $n ]
do
    flag=0
    for (( i=2; i<=num/2; i++ ))
    do
        if [ $((num % i)) -eq 0 ]; then
            flag=1
            break
        fi
    done
    if [ $flag -eq 0 ]; then
        echo -n "$num "
        count=$((count + 1))
    fi
    num=$((num + 1))
done
echo
Value for n :
5
Display of first n prime numbers
First 5 prime numbers are:
2 3 5 7 11

```

e) Write menu driven program for file handling activity

- Creation of file.
- Write content in the file.
- Upend file content.
- Delete file content.

CODE:

```
echo "1. Create file" echo "2.  
Write content to file" echo "3.  
Append content to file" echo  
"4. Delete file content" echo  
"Enter your choice:" read  
choice
```

```
echo "Enter filename:" read  
fname
```

```
case $choice in 1)  
touch $fname  
echo "File created"
```

```
;;
```

```
2) echo "Enter  
content:" cat >  
$fname
```

```
;;
```

```
3)
```

```
echo "Enter content to append:"  
cat >> $fname
```

```
;;
```

```
4)
```

```
> $fname  
echo "File content deleted"
```

```
;;
```

```
*) echo "Invalid  
choice"
```

```
;;
```

```
esac esac
```

OUTPUT:

```
hp@89L1D0M0F3O5T MINGW64 ~/Desktop/OSCD25D008 (main)
$ echo "1. Create file"
echo "2. Write content to file"
echo "3. Append content to file"
echo "4. Delete file content"
echo "Enter your choice:"
read choice
echo "Enter filename:"
read fname
case $choice in
1)
touch $fname
echo "File created"
;;
2)
echo "Enter content:"
cat > $fname
;;
3)
echo "Enter content to append:"
cat >> $fname
;;
4)
> $fname
echo "File content deleted"
;;
*)
echo "Invalid choice"
;;
esac
esac
1. Create file
2. Write content to file
3. Append content to file
4. Delete file content
Enter your choice:
2
Enter filename:
```